Water Quality Monitoring and Pool Maintenance System

Group ID: R26-IT-143

B.SC. (HONS) Degree in Information Technology Specialized in Information Technology

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TEAM



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Problem Background

- Swimming pools require continuous monitoring to maintain safety and hygiene.
- High swimmer density increases water contamination.
- Manual monitoring and fixed maintenance schedules are inefficient.
- Pool managers lack real-time insights to make maintenance decisions.

Real-World Challenges in Swimming Pool Monitoring

Users, Stakeholders and Impact of the Problem

Key Stakeholders

- Swimming Pool Managers
- Lifeguards
- Maintenance Staff
- Swimming Pool Visitors

Impact on Users and Society

- Poor water quality can cause infections and health risks
- Overcrowded pools increase safety concerns
- Manual monitoring increases operational workload
- Inefficient maintenance increases costs

Evidence

- WHO reports poorly maintained pools can cause waterborne diseases.

Existing Solutions and Their Limitations

Existing Solutions

- Manual monitoring by lifeguards
- Fixed maintenance schedules
- Basic water quality sensor monitoring

Limitations

- Cannot analyze swimmer crowd density
- Maintenance schedules are not adaptive
- Delayed detection of safety and hygiene issues
- Lack of integrated monitoring systems

Proposed Smart Pool Monitoring System

Solution Overview

- AI-based monitoring system for swimming pool management
- Combines video analysis and water quality sensors
- Provides real-time insights for maintenance decisions

[SYSTEM ARCHITECTURE DIAGRAM]

Main Components

- Swimmer Detection
- Crowd Density Estimation
- Water Quality Monitoring
- Maintenance Scheduling
- Pool Management Dashboard

Project Components and Individual Contributions

Student ID	Student ID	Student Name
IT22156310	M.H.M Saajith	Crowd-Aware Maintenance Scheduling
IT22156310	Yogarasa Nilogini	Water Quality Prediction
IT22248480	N.M Shafan	Water Quality Prediction
IT22277336	M.Y Shakeeb Arsalan	Garbage Detection

Crowd-Aware Maintenance Scheduling Module

Problem

- Maintenance schedules ignore real swimmer density
- High crowd levels accelerate water contamination
- Lack of intelligent monitoring systems

Objective

- Detect swimmers using CCTV cameras
- Estimate pool crowd density
- Generate dynamic maintenance schedules

Novelty

- AI-driven crowd-aware maintenance planning

Research Methodology – Water Quality Prediction

Data Collection

- Sensor data: pH, turbidity, chlorine, temperature

Data Preprocessing

- Data cleaning and normalization

Machine Learning Model

- Train classification model
- Predict Safe / Warning / Critical water conditions

Prediction and Analysis

- Real-time prediction of water quality

Technologies

- Python, Machine Learning, Data Analytics

Validation, Data Availability and Ethical Considerations

Data Availability

- Sensor data: pH, turbidity, chlorine, temperature
- Public water quality datasets for model training

Validation

- Evaluate machine learning model using accuracy metrics
- Compare predictions with real sensor measurements

Data Security

- Secure storage of sensor data
- Controlled system access

Ethical Considerations

- No personal data collection
- System designed to protect user safety and privacy

Implementation Overview – Water Quality Prediction Module

System Overview

- Sensor data collected from pH, turbidity, chlorine and temperature sensors
- Machine learning model predicts water quality levels
- Output classified as Safe, Warning or Critical

Dependencies

- Water Quality Sensor Module
- Maintenance Decision Engine
- Pool Management Dashboard

Commercialization Potential

Target Market

- Hotels and resorts
- Public swimming pools
- Sports complexes
- Pool management companies

Value Proposition

- AI-based water quality prediction
- Smart maintenance recommendations
- Reduced operational costs

Pricing Strategy

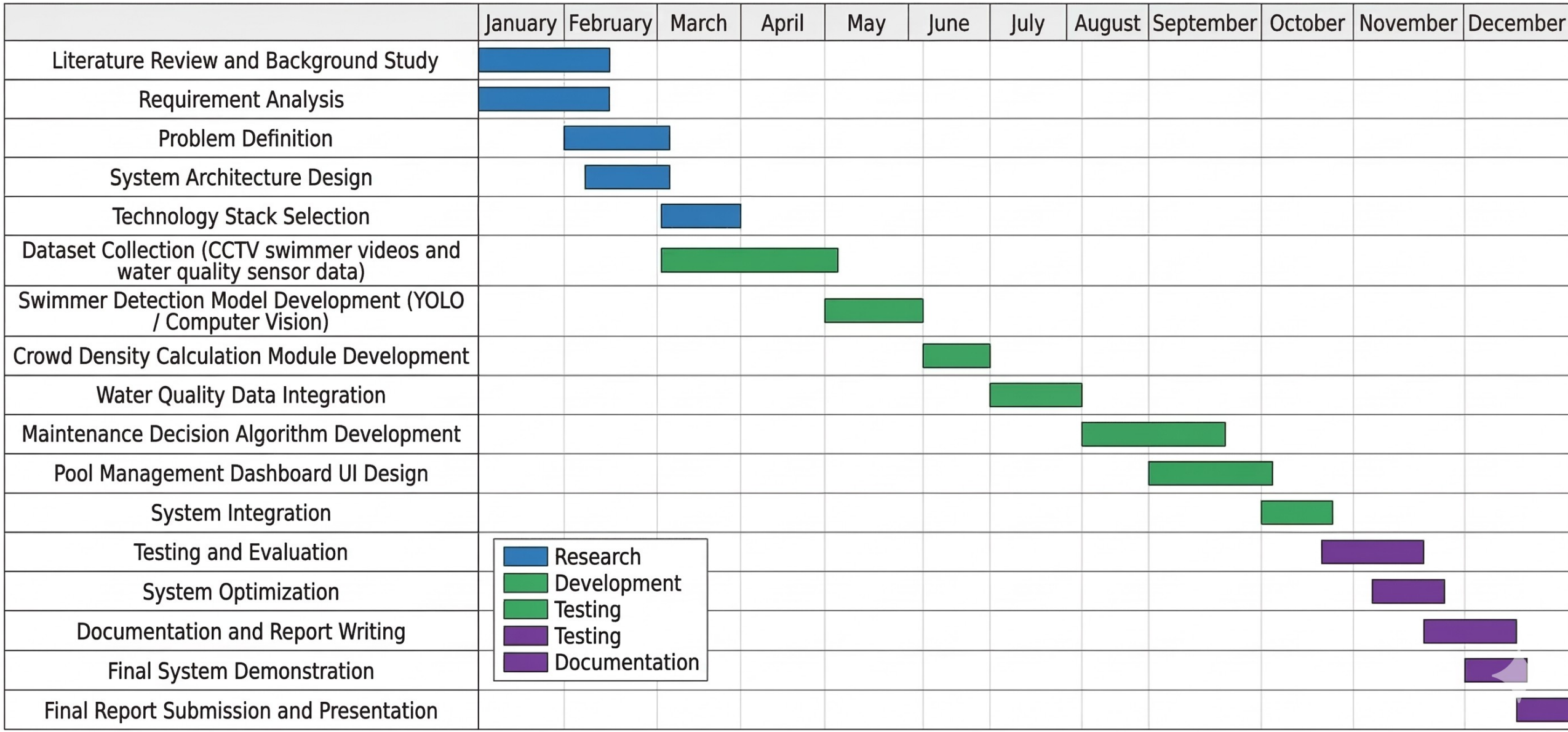
Basic – LKR 1,500/month
Standard – LKR 3,000/month
Premium – LKR 4,500/month

Competitive Advantage

- Integrated smart monitoring system
- Data-driven maintenance decisions

Crowd-Aware Pool Maintenance Scheduling System

Project Gantt Chart



Work Breakdown Structure (WBS)

Project Task Distribution

Crowd-Aware Maintenance Scheduling

- Analyze swimmer density
- Generate maintenance schedules

Water Quality Prediction

- Develop ML model for water quality classification

Drowning Detection

- Detect abnormal swimmer behavior

Garbage Detection

- Detect waste objects entering the pool